

MTH 302: Semantic Tableaux

October 28, 2025

α -rules (for branching):

α	α_1	α_2
$+\phi \wedge \psi$	$+\phi$	$+\psi$
$-\phi \vee \psi$	$-\phi$	$-\psi$
$-\phi \rightarrow \psi$	$+\phi$	$-\psi$
$+\neg\phi$	$-\phi$	
$-\neg\phi$	$+\phi$	

β -rules (non-branching):

β	β_1	β_2
$-\phi \wedge \psi$	$-\phi$	$-\psi$
$+\phi \vee \psi$	$+\phi$	$+\psi$
$+\phi \rightarrow \psi$	$-\phi$	$+\psi$

Quantifier-rules:

$+\exists x\phi(x)$	$+\phi(t)$, for any term t
$-\forall x\phi(x)$	$-\phi(t)$, for any term t
$+\forall x\phi(x)$	$+\phi(t)$, for any term t <i>not used in the expansion so far</i>
$-\exists x\phi(x)$	$-\phi(t)$, for any term t <i>not used in the expansion so far</i>

For x not free in ϕ :

$$\begin{array}{l|l} +\exists x\phi & +\phi \\ -\forall x\phi & -\phi \\ +\forall x\phi & +\phi \\ -\exists x\phi & -\phi \end{array}$$